

Purpose-specific minimization: 41.9% fewer tokens, zero identity values sent, accuracy preserved

Necessity Certificate benchmark on public data: fixed BANKING77-derived subset, 50 support queries (10 intents x 5), synthetic identity overlay, gpt-4o-mini, temperature 0, API-reported input tokens. Run 2026-09-03.

41.9%

fewer input tokens
(9,921 to 5,764)

300 to 0

raw identity values
sent to the model

88.9%

fewer fields sent
(450 to 50)

88% to 90%

accuracy: 1 error corrected,
0 regressions

Payload sent to the model	Accuracy	Input tokens	Token reduction	Fields sent	Identity values sent	Certificate
Full record	44/50 (88.0%)	9,921	0.0%	450	300	reference
Masking only	45/50 (90.0%)	9,371	5.5%	450	0	review (1 change)
Minimum context	45/50 (90.0%)	5,764	41.9%	50	0	review (1 change)
Minimum + masking	45/50 (90.0%)	5,764	41.9%	50	0	review (1 change)

Masking alone cuts 5.5% of tokens; deciding what never leaves the application cuts 41.9% (7.6x). The strict rule routes any changed decision, even an improvement, to human review. Synthetic support corpus (16 tickets): 44.9% fewer tokens, 96 PII values withheld, zero decision changes, accepted; the first candidate contract was rejected after one changed decision.

Limits: 50 cases over 10 of 77 intents, one run per condition, structured-field masker (not OpenAI Privacy Filter), public dataset with unknown pretraining exposure. Empirical evidence for an engineering review or a DPIA, not a legal proof. Source: BENCHMARK_RESULTS.md, benchmark-results/summary.csv (BANKING77, CC BY 4.0).